

Risk Factors for Severe Post-COVID Condition in Children, Adolescents, and Young Adults

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SUMMARY

Background

Post-COVID condition (PCC) in children and young people (PCCcyp) remains a significant health burden. Early identification of patients at risk for severe disease, including myalgic encephalomyelitis/ chronic fatigue syndrome (ME/CFS), is crucial to provide timely and adequate care.

Methods

This monocentric, observational registry study was performed at a tertiary pediatric hospital in Germany. Children and young people aged 7–25 years with post-COVID condition (PCC), according to the WHO, were included at the time of diagnosis. Standardized clinical assessment tools and patient-reported outcome measures were applied, including the novel Munich Long COVID Symptom Questionnaire (MLCSQ). Severe PCC was defined by clustered chronic symptom burden, fatigue severity scale (FSS), Total Composite Autonomic Symptom Score-31 (COMPASS-31), SF-36 composite scores, Bell Score, and confirmed ME/CFS diagnosis.

Findings

Of 120 participants, severe PCCcyp was associated with a higher number of acute symptoms (adjusted OR 1·22), acute orthostatic intolerance (adjusted OR 9·87), acute trouble concentrating (adjusted OR 11·8), and female sex (OR 3·31). Categorising acute symptoms at a threshold of ≥ 12 yielded the best model performance (AUC 0·857; sensitivity 65·6%; specificity 90·2%). ME/CFS was diagnosed in 24% participants, all within the severe PCC cluster, and was characterised by greater acute symptom complexity, more fatigue, more autonomic symptoms, and poorer physical function.

Interpretation

The number of acute symptoms and/or individual symptoms during the acute phase of the SARS-CoV-2 infection serve as early and specific predictors for severe PCCcyp. In particular, patients with ≥ 12 acute symptoms should be closely monitored to enable early diagnosis of severe PCC and potentially ME/CFS. A distinct cluster of severely affected patients – frequently with ME/CFS diagnosis – was identified.

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Research in context

Evidence before this study

Post-COVID condition (PCC) in children and young people (CYP) has been increasingly recognized, but robust evidence on predictors of severe disease has remained scarce. Previous reports in adult patients documented a wide spectrum of persistent symptoms, frequently including fatigue, neurocognitive complaints, and autonomic dysfunction, yet no study systematically investigated acute-phase predictors for severe PCCcyp.

Added value of this study

In this monocentric observational registry study on 120 CYP aged 7–25 years with PCC, we identified acute-phase risk factors for severe pediatric PCC. We show that a high number of acute symptoms (≥ 12 symptoms), female sex, acute orthostatic intolerance, and/or acute trouble concentrating strongly predicted a severe PCCcyp cluster. Importantly, 24% of patients fulfilled ME/CFS diagnostic criteria, and all clustered within the severe PCC group, underscoring the clinical overlap between severe pediatric PCC and ME/CFS.

Implications of all the available evidence

These findings provide a framework for early risk stratification in pediatric PCC. Recognising acute predictors is crucial to provide timely and adequate care, particularly to those with 12 or more acute symptoms and/or specific symptoms during the acute SARS-CoV-2 infection. The identification of a clinically relevant ME/CFS subgroup underscores the need to integrate ME/CFS expertise into pediatric PCC care pathways. Together, this evidence can inform clinical management, healthcare planning, and the design of interventional trials targeting the most vulnerable CYP with PCC.

INTRODUCTION

Post-acute sequelae of coronavirus disease 2019 (COVID-19) (PASC), often referred to as Long COVID, represent major clinical and economic challenges worldwide.¹⁻⁴ They include ongoing, relapsing, or new symptoms at four weeks or later after infection with severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2).⁵ A chronic form of PASC has been defined by the WHO as Post-COVID Condition (PCC). According to the definition for children and adolescents, PCC occurs in individuals with a history of confirmed or probable SARS-CoV-2 infection, when experiencing symptoms lasting at least 2 months which initially occurred within 3 months of acute COVID-19.

PCC can cause symptoms across all organ systems and often significantly impacts functioning, participation, and health-related quality of life (HRQoL).⁶ Frequent symptoms include fatigue, exertional intolerance, headache, respiratory symptoms, cognitive dysfunction, including brain fog, sleep problems, loss of smell and taste, anxiety, and autonomic dysfunction.^{7,8} The latter can manifest as orthostatic intolerance (OI) with or without postural orthostatic tachycardia syndrome (PoTS).⁹ Some patients develop post-exertional malaise (PEM), a compromising condition with symptom worsening after daily activities that were tolerated before. Long-lasting PEM is a cardinal symptom of myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS), a chronic multisystemic disease defined by clinical criteria and mostly triggered by viral infections.^{5,9,10} Most recent studies estimate ME/CFS prevalences ranging from 4.5 to 11.6% in adult PCC patients, depending on the study design.¹¹ Diagnosing PCC, PoTS, and ME/CFS requires a thorough diagnostic work-up to define clinical phenotypes and rule out other medical conditions with symptom overlap.

Data on PCC in CYP (PCCcyp) is still scarce, and age-specific factors are insufficiently understood.⁵ Depending on the study design, the reported PASC prevalence in minors ranged from 0.8% to 60%,¹² with lower rates observed in studies with SARS-CoV-2-negative control groups.^{6,12-15} While most CYP recover within a few months, a subset develops long-term disease.^{3,7,15,16}

Established risk factors for PCC in adults include infection with pre-Omicron variants,³ older age, female sex, pre-existing chronic conditions, lack of COVID-19 vaccination,¹⁷ more severe acute illness, and greater symptom complexity.¹⁸ Preliminary evidence suggests similar risk patterns in CYP,¹⁹⁻²² but risk factors of PCCcyp are more difficult to assess due to limited data and heterogenous age groups.

The IMMUC study on EBV-associated infectious mononucleosis (EBV-IM) and its long-term sequelae in CYP indicated that susceptibility to infection, a higher number of acute symptoms, and early gastrointestinal (GI) involvement predicted prolonged recovery and chronic fatigue.²³ Especially, gastrointestinal symptoms even prior to EBV primary infection has been identified as a risk factor for severe ME/CFS.

Here, we report on clinical phenotypes during the acute and chronic phases after SARS-CoV-2 infection of 120 CYP, including patients with post-COVID ME/CFS. We provide the first evidence for clinical risk factors of severe PCCcyp, which might facilitate timely identification and adequate care.

METHODS

Study Cohort and Case Definition

This monocenter observational registry study comprises medical data from 120 CYP aged 7 to 25 years and diagnosed with PCC according to WHO definitions²⁴ between January 2021 and May 2024 at the Munich Chronic Fatigue Center for Young People (MCFC).

Confirmed SARS-CoV-2 infection was defined by positive PCR or rapid antigen test combined with typical COVID-19 symptoms, while probable SARS-CoV-2 infection was defined by COVID-19 symptoms after contact with a SARS-CoV-2 PCR-positive person and positive SARS-CoV-2 serology not explained by vaccination.

Confirmed MECFS was defined by fulfilling the criteria of the Institute of Medicine (IOM) and/or the Canadian Consensus Criteria (CCC) as assessed by the Munich Berlin Symptom Questionnaire (MBSQ)⁹ in a structured medical interview after symptom-oriented differential diagnosis. In case of pediatric “pediatric case definition” of L.A. Jason and colleagues (PCD-J) and/or the clinical diagnostic worksheet designed by P.C. Rowe and colleagues (CDW-R)¹⁰ were used.

Assessed Data

Patients were asked to fill in questionnaires pre-admission, retrospectively addressing their and their family’s general medical history, their symptoms during the acute phase of the SARS-CoV-2 infection as well as their current symptoms, medical treatment, grade of participation in education/work, and HRQoL. To assess PCC symptoms, diagnostic ME/CFS criteria, severity of fatigue, presence, severity, and duration of PEM, autonomic symptoms, the daily activity level, and HRQoL we used the novel Munich Long COVID Symptom Questionnaire (MLCSQ), MBSQ⁹, the fatigue severity scale (FSS),²⁵ the DePaul Symptom Questionnaire for Post-exertional Malaise (DSQ-PEM),²⁶ the Composite Autonomic Symptom Score 31 (COMPASS 31),²⁷ the Bell CFIDS disability scale (Bell Score), and the Short Form-36 Health Survey (SF-36) with Physical (PCS) and Mental Component Summary (MCS),²⁸ respectively.

The MLCSQ was developed for structured age-independent assessment of PCC symptoms in 13 domains, encompassing 83 items (**Supplementary Material M1, Appendix pp.7-9**). Items were adapted from the WHO Global COVID-19 Clinical Platform Case Report Form (CRF) for PCC (2021), the Long COVID-19 Survey by the German Society for Pediatric Infectious Diseases (DGPI, 2021), and the COMPASS 31 (Sletten et al., 2012).²⁷ Frequency and severity were quantitatively assessed using a Likert scale as “not anymore/intermittent/persistent” or “mild/moderate/severe”, respectively. Only symptoms rated as “intermittent” or “persistent” in frequency were considered as PCC symptoms in this study.

Statistical Analysis

Descriptive Analysis: Continuous variables were described as median (interquartile range (IQR)) and compared between groups using Kruskal-Wallis or Wilcoxon Rank Sum tests. Categorical variables were described as absolute and relative frequencies (%) and tested with Chi-Squared or Fisher’s Exact tests as appropriate. A P-value < .05 was considered statistically significant.

Risk Factor Analysis Using Distinct Clinical Outcomes: Multivariable linear regression models for outcomes SF36-PCS, SF36-MCS, and the number of PCC symptoms, as well as logistic regression models for PEM, ME/CFS, and PoTS diagnosis, adjusted for sex and age, were used. Candidate predictor variables were COVID-19 vaccination status prior to infection,¹⁷ hospitalization during COVID-19,²⁹ the number of acute symptoms,²³ the presence of acute

gastrointestinal³⁰ or orthostatic intolerance symptoms, fatigue, and/or troubles concentrating during the acute SARS-CoV-2 infection.

Identification and Risk Factor Analysis for Severe PCC Cluster: A data-driven clustering approach was applied to identify patient subgroups based on PCC severity. The following outcome variables were used for the cluster definition: Bell Score, FSS, COMPASS 31 total score, SF36-PCS, SF36-MCS, number of PCC symptoms, ME/CFS diagnosis. A k-means clustering algorithm (k=2) was applied to the standardized data set. Two clusters encompassed patients with “moderate” and “severe” manifestations of these outcome variables, referred to as “moderate PCC” or “severe PCC”, respectively, for easy interpretability. The cluster membership was used as the primary outcome for subsequent analyses.

Using LASSO-penalized logistic regression, we assessed the relationship between features of the acute COVID-19 phase and subsequent cluster membership. Candidate predictor variables were the same as for the risk factor analysis using distinct clinical outcomes (see above).

The logistic LASSO regression model was fitted with 10-fold cross-validation to identify the optimal penalty parameter (λ) with the least deviance. To account for class imbalance between the “mild PCC” and “severe PCC” clusters, inverse class weights were applied.

The resulting coefficients of the cross-validated model can be used to calculate a predicted probability $p = \left(\frac{1}{1+e^{-z}} \right)$, for being attributed to the severe PCC cluster as, where $z = predictor_1 \times coefficient_1 + \dots + predictor_q \times coefficient_q$ and q the number of selected predictors by the LASSO operator. Model performance was evaluated using the area under the receiver operating characteristic curve (ROC; AUC), and the optimal classification threshold was derived using the Youden index.

Statistical analysis and data curation were performed using R, Version 4.2.1 “Funny Looking Kid” (The R Foundation for Statistical Computing, Vienna, Austria) and Python 3, Version 3.12.5 (Python Software Foundation, Delaware, USA). The Hartigan-Wong algorithm was used for the k-means clustering, implemented in the *clusters* R package. The *glmnet* and the *caret* packages were used for the LASSO-penalized logistic regression and cross-validation.

Ethics

The study adhered to the Declaration of Helsinki and all applicable regulations. The study protocol was approved by the *Ethikkommission der Fakultät für Medizin der Technischen Universität München* (Ethics Committee of the TUM University Hospital; approval 511/21, 2025-465-S-SB). Written informed consent was obtained from all participants aged 18–25 years; for participants aged 7–17 years, written informed consent was obtained from a parent or legal guardian, and age-appropriate assent was obtained from the participant. This observational registry study was registered at ClinicalTrials.gov (NCT05638724). No identifiable patient images or personal details are included in this Article; therefore, no additional publication permissions were required.

Role of the Funding Source

The funders had no involvement in the study’s design, data collection, analysis, interpretation, manuscript preparation, or the decision to submit the work for publication.

RESULTS

Patient Characteristics

120 CYP with PCC included 16/120 (13%) children (7-11 years), 71/12 (59%) adolescents (12-17 years), and 33/12 (28%) young adults (18-25 years). The percentage of females significantly increased with age (13% children, 62% adolescents, and 76% adults) (**Table 1**).

Characteristics of the Acute Phase of the SARS-CoV-2 Infection

Documented or probable SARS-CoV-2 infections occurred between February 2020 and January 2023, spanning pandemic waves in Germany with dominant Alpha (43/114 (38%)), Delta (25/114 (22%)), and Omicron variants (46/114 (40%)).

112/120 (93%) and 8/120 (7%) patients reported their first or second SARS-CoV-2 infection as PCC trigger, respectively. The triggering SARS-CoV-2 infection was indicated by PCR in 106/120 (88%), rapid antigen testing in 8/120 (7%), or serology together with contact in 6/120 (5%) CYP. 82/116 (71%) were not COVID-19 vaccinated prior to the index infection. A triggering breakthrough infection was reported in 34/116 (29%) patients (**Table 1**).

All patients remembered a median number of 16 (IQR: 11-21) acute COVID-19 symptoms during the index infection, with no significant difference between the three age groups ($P = 0.579$). The five most frequently reported COVID-19 symptoms were fatigue, headaches, sore throat, persistent dry cough, and trouble concentrating (**Table 1**). Loss of appetite ($P = 0.017$) and diarrhea ($P < 0.001$) were significantly more frequent in children. Heart racing ($P = 0.030$), loss of smell ($P < 0.001$) of taste ($P = 0.003$) were significantly more frequent in adolescents and young adults. Only 9/117 (8%) patients were hospitalized for their index infection (**Table 1**).

Characteristics of Post-COVID-Condition

Time from acute symptom onset to diagnosis was 8.2 month in median (IQR: 5.4 - 12.4) and significantly greater for young adults compare to children and adolescents ($P = 0.007$). Patients presented with various PCC-associated medical diagnoses (**Suppl. Table S1, Appendix p.2**). The three most frequent were fatigue (ICD-10-GM R53) (81/120, 68%), brain fog (ICD-10-GM F06.7) (64/120, 53%), and headaches (ICD-10-GM R51) (49/120, 41%). Brain fog (ICD-10-GM F06.7) was significantly more frequent in young adults compared to children and adolescents ($P = 0.034$).

The median number of PCC symptoms at the time of diagnosis was 29 (IQR: 21 – 37) and the most frequent symptoms were fatigue, exercise intolerance, reduced endurance, troubles in concentrating, and headaches, as indicated by the MLCSQ (**Table 2**). The median FSS score was 6.4 (IQR 5.8 to 6.6), reflecting pronounced fatigue (**Table 3**). The DSQ-PEM indicated PEM in 112/119 (94%) patients, with a duration of 14-23 or ≥ 24 hours reported by 19/114 (17%) and 23/114 (20%) patients, respectively (**Table 3**). The COMPASS 31 revealed a median score for OI of 20.0 (12.0 - 28.0) and a median total score of 27.8 (17.8 - 38.8), reflecting severe autonomic distress (**Table 3**).

The median Bell Score was 40% (IQR 30 to 60), indicating significant impairment of daily life (**Table 3**). 40/119 (34%) and 29/119 (24%) CYP stated to have less than 50% or no participation in school or work, respectively (**Table 3**). SF-36 median scores were markedly reduced in all eight dimensions compared to a German normative sample (aged 14–20 years, $N = 228$), with a median PCS of 31.0 (27.2 - 36.1) versus 57.1 (54.6 – 58.8) ($P < 0.025$) and a median MCS of 44.2 (35.1 - 51.0) versus 53.5 (50.4 – 55.9) ($P < 0.114$) (**Table 3**). PoTS (ICD-10-GM G90.80) was diagnosed in 25/120 (21%) patients (**Table 3**).

Sub-Group Comparison of ME/CFS vs. non-ME/CFS Patients

The diagnosis of ME/CFS (ICD-10-GM G93.3) was confirmed or probable after medical assessment in 29/120 (24.2%) patients (**Table 3**) and was more frequent with older age (children 19%, adolescents 20%, adults 36%).

Confirmed or probable ME/CFS compared to non-ME/CFS patients had a greater number of acute COVID-19 symptoms (median 17.5 (14.2 – 20.0) versus 14.0 (10.0 – 18.0); $P = 0.021$), a greater number of PCC symptoms (median 18.0 (16.0 – 22.0) versus 16.0 (11.5 – 20.0); $P = 0.032$), higher FSS scores (median 6.6 (6.3 – 6.8) versus 6.2 (5.7 – 6.6); $P = 0.012$), higher proportion of PEM duration ≥ 24 h: 37.9% versus 14.0%, higher COMPASS 31 total scores (median 34.9 (25.6 – 43.5) versus 25.6 (16.8 – 36.5); $P = 0.032$), lower Bell Scores (median 30.0 (30.0 – 50.0) versus 50.0 (40.0 – 60.0); $P < 0.001$), and lower scores of SF36-PCS (median 27.1 (23.6 – 32.0) versus 31.5 (28.3 – 38.6); $P = 0.025$) and SF36-MCS (median 42.4 (29.5 – 47.9) versus 45.6 (36.0 – 52.7), $P = 0.114$). (**Suppl. Table S1, Appendix p.2**)

Risk Factors Analysis Using Distinct Clinical Outcomes

Possible associations of individual candidate predictor variables during the acute phase of the SARS-CoV-2 infection and distinct outcomes at the time of PCC diagnoses were investigated:

- i) *Acute OI* was significantly associated with a higher FSS score (adj. coefficient: 0.377, $P = 0.042$), a higher COMPASS 31 total score (adj. coefficient: 12.570, $P < 0.001$), and a lower SF36-MCS (adj. coefficient: -6.185, $P = 0.008$);
- ii) *the presence of OI or GI symptoms* was significantly associated with a higher number of PCC symptoms (OI: adj. coefficient: 9.276, $P < 0.001$; GI: adj. coefficient: 9.668, $P < 0.001$);
- iii) *hospitalization due to COVID-19* was associated with a lower Bell score (adj. coefficient: -8.991, $P = 0.176$) and a lower SF36-MCS (adj. coefficient: -7.685, $P = 0.048$);
- iv) *a higher number of COVID-19 symptoms* was significantly associated with higher FSS scores (adj. coefficient: 0.045, $P < 0.001$), a higher COMPASS 31 total score (adj. coefficient: 0.633, $P = 0.010$), a lower SF36-PCS (adj. coefficient: -0.522, $P = 0.001$), a lower SF36-MCS (adj. coefficient: -0.335, $P = 0.060$), and a higher number of chronic symptoms (adj. coefficient: 1.142, $P < 0.001$);
- v) *acute trouble concentrating* were significantly associated with a higher FSS score (adj. coefficient: 0.502, $P = 0.013$), a higher SF36-PCS (adj. coefficient: -6.232, $P = 0.007$), a higher number of chronic symptoms (adj. coefficient: 10.9, $P < 0.001$) and a higher likelihood for a positive DSQ-PEM screening (adj. OR: 2.360, $P = 0.019$);
- vi) *older age* was significantly associated with a higher likelihood for positive PEM as indicated by the DSQ-PEM (adj. OR: 1.509, $P = 0.009$), independent of sex, and
- vii) *being female* was associated with a significantly higher likelihood of confirmed or suspected medical ME/CFS diagnosis (G93.3) and a significantly higher COMPASS 31 total score (adj. coefficient: 9.657, $P = 0.005$). None of the acute features were significantly associated with brain fog (F06.7). Results are summarized in **Table 4**.

Risk Factors Analysis for Severe PCC

The result of the k-means clustering is visualized in **Suppl. Figure S1A (Appendix p.6)**. The first cluster encompassed all 29/70 (41%) patients with confirmed or probable ME/CFS and was further characterized by a lower Bell Score, more severe fatigue (FSS), more autonomic symptoms (COMPASS 31), more PCC symptoms, a lower SF36-PCS and a lower SF36-MCS compared to the second cluster (**Suppl. Figure S1B-S1H**). Female sex, but not age, had a significantly greater likelihood of being attributed to the severe PCC cluster (OR = 3.31, P = 0.031). Adjusted for sex and age, the number of acute symptoms (adj. OR = 1.22, P < 0.001), acute symptoms of OI (adj. OR = 9.87, P = 0.002), and acute troubles concentrating (adj. OR = 11.8, P = 0.005) were significantly associated with the severe PCC cluster. (**Figure 1**)

A trained and cross-validated LASSO retained the following predictors a final logistic regression model: female sex (coefficient: 1.08), age (coefficient: -0.07), hospitalization during acute COVID-19 (coefficient: 0.10), number of acute COVID-19 symptoms (coefficient: 0.10), acute OI symptoms (coefficient: 1.09), acute fatigue (coefficient: -0.95), and acute troubles concentrating (coefficient: 0.80). A ROC analysis of this model resulted in an AUC of 85.7%. The optimal threshold for predicting severe PCC was if the predicted probability $p \geq 63.8\%$ (Youden index; **Suppl. Figure S1I**). The model reliably detected CYP with “severe PCC” (sensitivity: 65.6%) or “moderate PCC” (specificity: 90.2%), yielding an overall accuracy of 80.0% (balanced accuracy: 77.9%) (**Suppl. Figure S1J, S1K**). A binary categorization of the acute number of symptoms in the penalized logistic regression model revealed that ≥ 12 acute COVID-19 symptoms had the best model fit, indicated by minimal cross-validated model error (**Suppl. Figure S1L**).

DISCUSSION

This study identified key risk factors for severe PCC in CYP after confirmed or probable COVID-19, highlighted a distinct subgroup of highly affected individuals, and introduced the novel MLCSQ as a robust instrument for standardized assessment of PCC symptoms. The most important finding was that a high number of acute symptoms and the presence of autonomic symptoms during the acute phase of initial SARS-CoV-2 infection were early and specific predictors of severe PCC and thereby may allow for tailored and timely symptom-oriented care.

A higher number of acute initial symptoms was significantly associated with more severe long-term outcomes, including more severe fatigue (FSS) and autonomic dysfunction (COMPASS-31), reduced physical and mental HRQoL (SF-36), and a higher number of chronic symptoms (MLCSQ). These results expand on earlier work, which indicated that acute EBV-IM symptom complexity and autonomic EBV-IM symptoms predicted prolonged recovery and/or chronic fatigue.²³ The latter finding was consistent with results from Jason et al., who identified greater symptom burden and immune activation as predictors of EBV-IM-driven ME/CFS.

In adults, the presence of five acute COVID-19 symptoms has been associated with increased PCC risk.¹⁸ Moreover, symptomatic versus asymptomatic SARS-CoV-2 infection has been linked to long-term sequelae in adults and children. While earlier pediatric studies identified hospitalization for COVID-19^{3,19,20} and acute complications such as pneumonia, reduced oxygen saturation, and hypotension as risk factors for PCC, our data are the first to demonstrate that a threshold of ≥ 12 COVID-19 symptoms allows for identifying CYP at risk of severe PCC.

Among acute COVID-19 symptoms, symptoms of OI (dizziness, syncope, blackouts, and circulatory dysregulation) emerged as another strong predictor of severe PCC. These symptoms were significantly associated with increased fatigue, greater autonomic distress, poorer mental health, and a higher number of symptoms in the context of PCC. This supports the hypothesis that autonomic dysregulation plays a key role in the development of post-viral sequelae.¹⁴

Unsupervised clustering delineated a clinically coherent “severe” phenotype that concentrated all ME/CFS cases and showed markedly worse fatigue, autonomic symptom burden, functional status, and HRQoL than the comparison cluster. This supports the robustness of our cluster endpoint.

The risk-factor analyses found that female sex was associated with higher odds of severe PCC, which is consistent with prior pediatric reports.⁸ In contrast to several pediatric cohort studies reporting, we could not confirm age as a relevant risk factor for PCC.³ E.g., Rao et al. found that children younger than five years were more strongly associated with PASC.¹⁹ This controversy might indicate that more nuanced age-specific modelling is needed. Acute orthostatic-intolerance symptoms and acute troubles concentrating exhibited the strongest adjusted associations with severe PCC, highlighting early autonomic and neurocognitive involvement as salient signals. A higher acute symptom burden also tracked with risk, with the penalized model indicating ≥ 12 acute symptoms as a pragmatic cut-point. It is not surprising that acute fatigue was not identified as a discriminatory risk factor since it was prevalent in 90% of all patients.

LASSO-penalized logistic regression selected seven independent predictors for severe PCC, including female sex, younger age, hospitalization, number of acute symptoms, acute OI symptoms, acute troubles concentrating, and absence of acute fatigue. In accordance, female sex has been considered a risk factor in pediatric patients.⁸ Interestingly, acute fatigue and sex now appeared as risk factors. However, they may merely reflect an additional effect in the

presence of the remaining information of the other predictors in the model. This model surely needs further evaluation in further studies.

All ME/CFS patients included in the cluster analysis were assigned to the severe PCC cluster. This subgroup of ME/CFS patients was characterized by older age, female predominance, higher acute symptom burden, and more severe scores across all domains. This aligns with previous studies reporting that ME/CFS is often associated with OI in adolescents,¹⁰ and that pediatric PCC patients with ME/CFS, compared to non-ME/CFS, had a higher symptom load.⁵

Strengths and Limitations

A key strength of our work is its clinical immediacy: we identify a pragmatic threshold, i.e., ≥ 12 acute symptoms, that enables early risk stratification for severe PCC in CYP. The study leverages a relatively large, well-characterized pediatric cohort and a comprehensive, multidimensional assessment using validated clinical instruments and PROMs, enhancing reproducibility and comparability. Our risk model is robust and data-driven, combining unsupervised clustering with LASSO and cross-validation to prioritize predictors with joint explanatory value. Finally, the retained predictors (sex, age, hospitalization, acute symptom load, acute OI symptoms, acute concentration problems, and acute fatigue) are easy to assess, supporting direct clinical translation.

Unfortunately, this analysis did not include a control group that fully recovered after SARS-CoV-2 infection. Therefore, risk factors were conditional on PCC development, and general extrapolation to CYP with SARS-CoV-2 infections remains limited. Moreover, COVID-19 symptoms might be influenced by recall bias, and the proposed clinical cut-off of ≥ 12 acute symptoms requires further validation, including age- and sex-specific analyses in larger, independent cohorts. Additional limitations include missing data in some outcome variables, which reduced the effective sample size for clustering and regression analyses and may have affected the stability of model estimates. Lastly, findings for our German monocenter cohort may not be generalizable to other populations or healthcare systems.

This study highlights the importance of acute-phase symptom profiles, particularly the number of symptoms and autonomic dysfunction, for identifying CYP at risk for severe PCC. The novel MLCSQ enables a standardized, quantitative assessment of PCC symptom load. Our findings emphasize the need for systematic early symptom documentation to guide timely risk stratification, individualized follow-up, and targeted intervention in pediatric populations affected by long-term sequelae of COVID-19.

Contributors

Q.D., Ma.Hä., D.S., C.C., M.A., LC.P., S.S., K.G., R.P., L.M., and U.B. conceptualized the study and developed the methodology. Data acquisition and curation were performed by Q.D., Ma.Hä., D.S., T.W., C.C., A.Gr., A.L., C.I., Ca.W., M.E., S.B., N.B., H.W., H.H., LC.P., L.B., S.S., Co.Wa., M.A., Ar.Gr., R.P., and U.B.. Formal analyses were conducted by Q.D., L.M., and U.B.. All authors contributed to data validation of the study. Supervision was provided by U.B. and L.M., with project administration and funding acquisition by U.B. Software and visualisation were handled by Q.D. and L.M.. The original draft was written by Q.D., L.M., and U.B., and all authors reviewed and edited the manuscript critically. U.B., L.M., and Q.D. have directly accessed and verified all underlying data. All authors had full access to all study data, accept responsibility for the decision to submit for publication, and agree to be accountable for all aspects of the work.

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Declaration of interests

All authors have completed the ICMJE uniform disclosure form. U.B. reports research funding from the Bavarian State Ministry of Health and Care, the Bavarian State Ministry of Science and Art, the German Center for Infection Research (DZIF), the Federal Ministry of Research, Technology and Space, the Federal Ministry of Health, and the Federal Joint Committee. CS received grants from the Federal Ministry of Health, consulting fees from Celltrend, and payment or honoraria from Bayer, Fresenius, Roche, Celltrend, Hausärzteverband Berlin, AGBAN AKADEMIE, and Berufsverband Dt. Internisten. Further, CS is a participant in advisory boards at Berlin cures and the Federal Institute for Drugs and Medical Devices (BfArM). M.A. received financial compensation from BioNTech and Pfizer for participating in advisory board meetings on Long COVID. C.C. received personal payment from the Austrian Chamber of Pharmacists for delivering vaccination training lectures. Q.D. and LC.P. report an MD scholarship from the German Center for Infection Research. LC.P. received a travel scholarship from the Lost Voices Foundation to attend the 2023 IACFS/ME Research and Clinical Conference. N.T. serves as Vice President of the German Society of Pediatric Infectious Diseases (DGPI). U.B. and C.S. are members of the medical advisory board of the German association for ME/CFS. All other authors declare no competing interests.

Data sharing statement

All data supporting the findings of this study is reported in this published manuscript and its supplementary Appendix. Ethics approval and patient consent forms do not allow individual participant data to be made publicly available. Requests for further information can be directed to the corresponding author.

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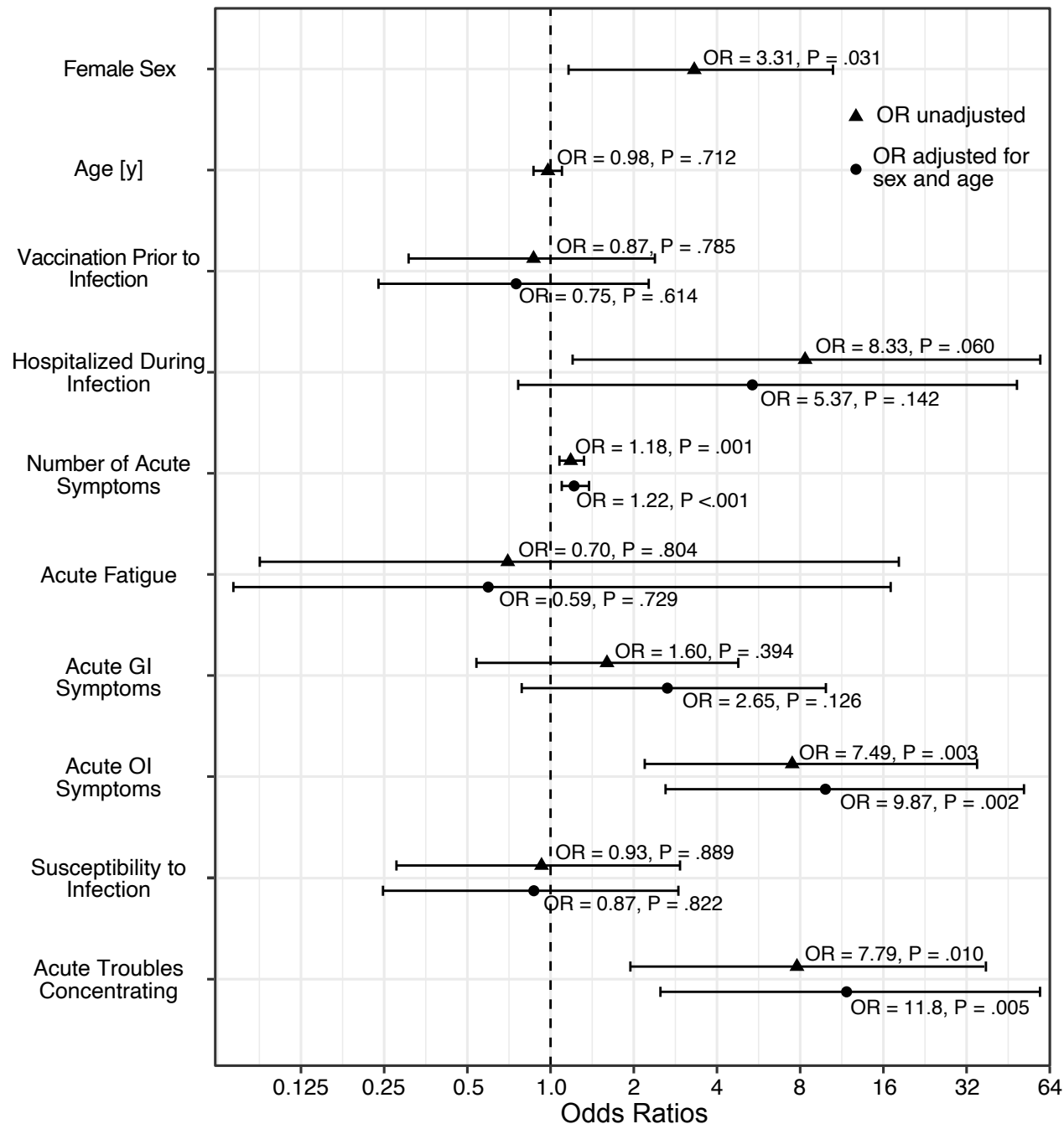
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FIGURE LEGEND

Figure 1. Forest plot of predictors for severe vs. moderate PCC_{cyp} ($N=70$) derived from logistic regression models. Circles show unadjusted odds ratios (OR), triangles show OR adjusted for sex and age. Bars indicate 95%-confidence intervals.

Risk Factors for Severe PCC



TABLES

Table 1. Patient's Basic Characteristics, Vaccination Status, and Symptoms During the Acute Phase of PCC-related SARS-CoV-2 Infection.

	Overall^a	Children^a	Adolescents^a	Young Adults^a	P-Value^b
	N = 120	Age 7 to 11 years N = 16	Age 12 to 17 years N = 71	Age 18 to 25 years N = 33	
Age ^a , years	15·0 (13·0 - 18·0)	11·0 (9·8 - 11·0)	15·0 (13·0 - 16·0)	21·0 (19·0 - 23·0)	NA
Sex, female	71 / 120 (59%)	2 / 16 (13%)	44 / 71 (62%)	25 / 33 (76%)	<0·001
Body height ^a , cm	167·0 (160·8 - 173·8)	146·0 (141·8 - 152·3)	167·0 (161·8 - 173·3)	170·1 (166·5 - 177·0)	NA
Body weight ^a , kg	58·5 (48·4 - 68·6)	35·0 (31·7 - 45·8)	60·7 (50·9 - 68·6)	62·5 (56·6 - 75·0)	NA
Body mass index ^a , kg/m ²	20·7 (18·4 - 22·9)	17·1 (15·9 - 19·0)	21·3 (18·8 - 22·9)	21·1 (19·5 - 24·7)	<0·001
Allergies	51 / 116 (44%)	6 / 15 (40%)	29 / 69 (42%)	16 / 32 (50%)	0·714
Allergies since PCC onset	4 / 39 (10%)	1 / 2 (50%)	2 / 22 (9·1%)	1 / 15 (6·7%)	0·214
Susceptibility to infection ^c	26 / 110 (24%)	2 / 14 (14%)	15 / 65 (23%)	9 / 31 (29%)	0·614
Standard vaccination according to STIKO	93 / 95 (98%)	12 / 13 (92%)	59 / 60 (98%)	22 / 22 (100%)	0·308
COVID-19 vaccinations before symptom onset	NA	NA	NA	NA	0·007
none	82 / 116 (71%)	14 / 16 (88%)	40 / 69 (58%)	28 / 31 (90%)	NA
1	0 / 116 (0%)	0 / 16 (0%)	0 / 69 (0%)	0 / 31 (0%)	NA
2	26 / 116 (22%)	2 / 16 (13%)	21 / 69 (30%)	3 / 31 (10%)	NA
3	8 / 116 (7%)	0 / 16 (0%)	8 / 69 (12%)	0 / 31 (0%)	NA
PCC-triggering SARS-CoV-2 infection	NA	NA	NA	NA	0·430
1st	112 / 120 (93%)	14 / 16 (88%)	66 / 71 (93%)	32 / 33 (97%)	NA
2nd	8 / 120 (7%)	2 / 16 (13%)	5 / 71 (7%)	1 / 33 (3%)	NA
Documentation of SARS-CoV-2 infection	NA	NA	NA	NA	0·166
PCR	106 / 120 (88%)	14 / 16 (88%)	61 / 71 (86%)	31 / 33 (94%)	NA
Rapid antigen test	8 / 120 (7%)	2 / 16 (13%)	6 / 71 (9%)	0 / 33 (0%)	NA
Serology	6 / 120 (5%)	0 / 16 (0%)	4 / 71 (6%)	2 / 33 (6%)	NA

Time from infection to PCC symptom onset, days	0·5 (-0·3 - 30·8)	0·0 (-0·3 - 29·5)	12·0 (.0 - 38·5)	0·0 (-3·0 - 8·0)	0·005
Time from infection to diagnosis, months	8·2 (5·4 - 12·4)	6·5 (5·6 - 8·8)	7·5 (5·0 - 12·1)	10·8 (7·9 - 14·5)	0·007
Acute symptom treatment	NA	NA	NA	NA	0·806
No medical visit	85 / 117 (73%)	12 / 16 (75%)	48 / 68 (71%)	25 / 33 (76%)	NA
Outpatient	23 / 117 (20%)	4 / 16 (25%)	13 / 68 (19%)	6 / 33 (18%)	NA
Hospitalized	9 / 117 (8%)	0 / 16 (0%)	7 / 68 (10%)	2 / 33 (6%)	NA
Number of acute symptoms reported	16·0 (11·0 - 21·0)	17·0 (13·0 - 18·5)	14·5 (11·0 - 20·0)	16·0 (12·0 - 23·0)	0·580
Acute Symptoms	NA	NA	NA	NA	NA
Fatigue	104 / 116 (90%)	15 / 15 (100%)	60 / 68 (88%)	29 / 33 (88%)	0·503
Headaches	103 / 116 (89%)	13 / 15 (87%)	60 / 68 (88%)	30 / 33 (91%)	0·918
Sore throat	91 / 116 (78%)	14 / 15 (93%)	55 / 68 (81%)	22 / 33 (67%)	0·107
Persistent dry cough	87 / 116 (75%)	11 / 15 (73%)	54 / 68 (79%)	22 / 33 (67%)	0·375
Trouble in concentrating	85 / 116 (73%)	13 / 15 (87%)	47 / 68 (69%)	25 / 33 (76%)	0·415
Shortness of breath	84 / 116 (72%)	11 / 15 (73%)	50 / 68 (74%)	23 / 33 (70%)	0·955
Cold	83 / 116 (72%)	9 / 15 (60%)	53 / 68 (78%)	21 / 33 (64%)	0·184
Dizziness	75 / 116 (65%)	9 / 15 (60%)	43 / 68 (63%)	23 / 33 (70%)	0·752
Muscle pain	70 / 116 (60%)	8 / 15 (53%)	43 / 68 (63%)	19 / 33 (58%)	0·722
Fever	65 / 116 (56%)	11 / 15 (73%)	38 / 68 (56%)	16 / 33 (48%)	0·274
Loss of appetite	61 / 116 (53%)	13 / 15 (87%)	33 / 68 (49%)	15 / 33 (45%)	0·017
Chest tightness	59 / 116 (51%)	9 / 15 (60%)	30 / 68 (44%)	20 / 33 (61%)	0·224
Joint pain	59 / 116 (51%)	7 / 15 (47%)	34 / 68 (50%)	18 / 33 (55%)	0·859
Nausea	53 / 116 (46%)	8 / 15 (53%)	34 / 68 (50%)	11 / 33 (33%)	0·235
Stomach pain	51 / 116 (44%)	10 / 15 (67%)	28 / 68 (41%)	13 / 33 (39%)	0·163
Pain on breathing	47 / 116 (41%)	6 / 15 (40%)	24 / 68 (35%)	17 / 33 (52%)	0·297
Palpitations	45 / 116 (39%)	6 / 15 (40%)	23 / 68 (34%)	16 / 33 (48%)	0·364
Tachykardia	45 / 116 (39%)	4 / 15 (27%)	22 / 68 (32%)	19 / 33 (58%)	0·030
Swollen lymph nodes	44 / 116 (38%)	8 / 15 (53%)	27 / 68 (40%)	9 / 33 (27%)	0·202
Loss of smell	44 / 116 (38%)	0 / 15 (0%)	25 / 68 (37%)	19 / 33 (58%)	<0·001
Apathy	43 / 116 (37%)	5 / 15 (33%)	22 / 68 (32%)	16 / 33 (48%)	0·275

Light-headedness	42 / 116 (36%)	4 / 15 (27%)	24 / 68 (35%)	14 / 33 (42%)	0.558
Weight loss	38 / 116 (33%)	5 / 15 (33%)	21 / 68 (31%)	12 / 33 (36%)	0.883
Loss of taste	38 / 116 (33%)	0 / 15 (0%)	23 / 68 (34%)	15 / 33 (45%)	0.003
Altered taste	32 / 116 (28%)	4 / 15 (27%)	19 / 68 (28%)	9 / 33 (27%)	>0.999
Altered smell	29 / 116 (25%)	3 / 15 (20%)	17 / 68 (25%)	9 / 33 (27%)	0.909
Breath-independent chest pain	28 / 116 (24%)	6 / 15 (40%)	12 / 68 (18%)	10 / 33 (30%)	0.100
Numbness or tingling	26 / 116 (22%)	5 / 15 (33%)	14 / 68 (21%)	7 / 33 (21%)	0.561
Diarrhoea	25 / 116 (22%)	9 / 15 (60%)	9 / 68 (13%)	7 / 33 (21%)	<0.001
Fainting/blackouts	14 / 116 (12%)	1 / 15 (7%)	11 / 68 (16%)	2 / 33 (6%)	0.393
Vomiting	13 / 116 (11%)	2 / 15 (13%)	7 / 68 (10%)	4 / 33 (12%)	0.841
Conjunctivitis	11 / 116 (10%)	0 / 15 (0%)	7 / 68 (10%)	4 / 33 (12%)	0.483
Skin rash	9 / 116 (8%)	2 / 15 (13%)	3 / 68 (4%)	4 / 33 (12%)	0.226
Pain in the limbs	7 / 116 (6%)	0 / 15 (0%)	3 / 68 (4%)	4 / 33 (12%)	0.252
COVID-19 toes (purple, pink, bluish)	6 / 116 (5%)	0 / 15 (0%)	5 / 68 (7%)	1 / 33 (3%)	0.585

a Data are expressed as n/N (%), median (IQR), or mean (SD), as appropriate. Percentages may not total 100% because of rounding.

b P-values were calculated using the χ^2 test, Fisher exact test, or Kruskal-Wallis test, as appropriate.

c Susceptibility to infection was defined as ≥ 1 of the following: recurrent infections at the same site, >2 antibiotic courses/year, hospitalization due to infection, ≥ 4 episodes of herpes labialis/year, at least one episode of genital herpes or herpes zoster.

d Abbreviations: PCR, polymerase chain reaction; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2; STIKO, Standing Committee on Vaccination at the Robert Koch Institute (Germany).

NA = not available.

Table 2. *Number of Chronic Postviral Symptoms Assessed by the Munich Long COVID Symptom Questionnaire (MLCSQ).*

	Overall^a	Children^a	Adolescents^a	Young Adults^a	P-Value^b
	N = 120	Age 7 to 11 years N = 16	Age 12 to 17 years N = 71	Age 18 to 25 years N = 33	
Number of 83 possible PCC symptoms reported	29.0 (21.0 - 37.0)	29.5 (18.0 - 35.25)	29.0 (20.0 - 36.0)	31.0 (24.5 - 37.75)	0.448
Fatigue / Exercise intolerance	NA	NA	NA	NA	NA
Fatigue	105 / 109 (96%)	14 / 16 (88%)	63 / 65 (97%)	28 / 28 (100%)	0.097
Worsening of symptoms following mild mental and/or physical activity (exercise intolerance)	104 / 109 (95%)	14 / 16 (88%)	62 / 65 (95%)	28 / 28 (100%)	0.162
Reduced endurance	103 / 109 (94%)	14 / 16 (88%)	61 / 65 (94%)	28 / 28 (100%)	0.203
Neurological	NA	NA	NA	NA	NA
Trouble in concentrating	97 / 109 (89%)	15 / 16 (94%)	55 / 65 (85%)	27 / 28 (96%)	0.200
Headaches	96 / 109 (88%)	14 / 16 (88%)	56 / 65 (86%)	26 / 28 (93%)	0.656
Dizziness	84 / 109 (77%)	10 / 16 (62%)	54 / 65 (83%)	20 / 28 (71%)	0.153
Forgetfulness	81 / 109 (74%)	11 / 16 (69%)	47 / 65 (72%)	23 / 28 (82%)	0.523
Light-headedness	54 / 109 (50%)	3 / 16 (19%)	33 / 65 (51%)	18 / 28 (64%)	0.014
Problems with balance	50 / 109 (46%)	6 / 16 (38%)	33 / 65 (51%)	11 / 28 (39%)	0.456
Numbness or tingling	34 / 109 (31%)	7 / 16 (44%)	17 / 65 (26%)	10 / 28 (36%)	0.331
Slowness of movement	28 / 109 (26%)	5 / 16 (31%)	16 / 65 (25%)	7 / 28 (25%)	0.858
Problems with gait/falls	21 / 109 (19%)	4 / 16 (25%)	16 / 65 (25%)	1 / 28 (4%)	0.051
Seizures	8 / 109 (7%)	1 / 16 (6%)	6 / 65 (9%)	1 / 28 (4%)	0.621
Can't feel one side of body or face	7 / 109 (6%)	0 / 16 (0%)	6 / 65 (9%)	1 / 28 (4%)	0.312
Can't move one side of body or face	1 / 109 (1%)	0 / 16 (0%)	0 / 65 (0%)	1 / 28 (4%)	0.232
Pulmonary	NA	NA	NA	NA	NA
Shortness of breath with activity	88 / 109 (81%)	12 / 16 (75%)	54 / 65 (83%)	22 / 28 (79%)	0.722
Persistent dry cough	48 / 109 (44%)	7 / 16 (44%)	30 / 65 (46%)	11 / 28 (39%)	0.829
Shortness of breath at rest	44 / 109 (40%)	4 / 16 (25%)	30 / 65 (46%)	10 / 28 (36%)	0.256
Pain on breathing	43 / 109 (39%)	3 / 16 (19%)	24 / 65 (37%)	16 / 28 (57%)	0.035
Congestion	36 / 109 (33%)	2 / 16 (12%)	24 / 65 (37%)	10 / 28 (36%)	0.167

Cardiovascular	NA	NA	NA	NA	NA
Orthostatic intolerance	83 / 109 (76%)	6 / 16 (38%)	53 / 65 (82%)	24 / 28 (86%)	<0.001
Heart pounding	55 / 109 (50%)	6 / 16 (38%)	31 / 65 (48%)	18 / 28 (64%)	0.181
Chest tightness	54 / 109 (50%)	4 / 16 (25%)	30 / 65 (46%)	20 / 28 (71%)	0.009
Heart racing	51 / 109 (47%)	5 / 16 (31%)	30 / 65 (46%)	16 / 28 (57%)	0.251
Breath-independent chest pain	25 / 109 (23%)	3 / 16 (19%)	13 / 65 (20%)	9 / 28 (32%)	0.403
Fainting and/or Blackouts	16 / 109 (15%)	1 / 16 (6%)	15 / 65 (23%)	0 / 28 (0%)	0.009
Sleep	NA	NA	NA	NA	NA
Sleeping more	81 / 109 (74%)	10 / 16 (62%)	47 / 65 (72%)	24 / 28 (86%)	0.201
Other sleep disturbances	50 / 109 (46%)	7 / 16 (44%)	30 / 65 (46%)	13 / 28 (46%)	0.983
Sleeping less	37 / 109 (34%)	3 / 16 (19%)	22 / 65 (34%)	12 / 28 (43%)	0.267
Mental health	NA	NA	NA	NA	NA
Listlessness / Lack of motivation	80 / 109 (73%)	11 / 16 (69%)	50 / 65 (77%)	19 / 28 (68%)	0.597
Loss of interest/pleasure	61 / 109 (56%)	7 / 16 (44%)	37 / 65 (57%)	17 / 28 (61%)	0.535
Depressed mood	59 / 109 (54%)	5 / 16 (31%)	37 / 65 (57%)	17 / 28 (61%)	0.131
Anxiety	44 / 109 (40%)	7 / 16 (44%)	25 / 65 (38%)	12 / 28 (43%)	0.884
Behavioural change	35 / 109 (32%)	5 / 16 (31%)	20 / 65 (31%)	10 / 28 (36%)	0.893
Tic	11 / 109 (10%)	2 / 16 (12%)	8 / 65 (12%)	1 / 28 (4%)	0.414
Hallucinations	6 / 109 (6%)	1 / 16 (6%)	5 / 65 (8%)	0 / 28 (0%)	0.325
Musculoskeletal	NA	NA	NA	NA	NA
Muscle pain	69 / 109 (63%)	12 / 16 (75%)	38 / 65 (58%)	19 / 28 (68%)	0.397
Muscle weakness	63 / 109 (58%)	9 / 16 (56%)	35 / 65 (54%)	19 / 28 (68%)	0.451
Back pain	59 / 109 (54%)	9 / 16 (56%)	33 / 65 (51%)	17 / 28 (61%)	0.666
Joint pain	52 / 109 (48%)	9 / 16 (56%)	29 / 65 (45%)	14 / 28 (50%)	0.678
Muscle twitching	36 / 109 (33%)	4 / 16 (25%)	21 / 65 (32%)	11 / 28 (39%)	0.614
Muscle tremor	34 / 109 (31%)	3 / 16 (19%)	23 / 65 (35%)	8 / 28 (29%)	0.411
Muscle stiffness	17 / 109 (16%)	3 / 16 (19%)	7 / 65 (11%)	7 / 28 (25%)	0.207
Joint swelling	5 / 109 (5%)	2 / 16 (12%)	2 / 65 (3%)	1 / 28 (4%)	0.260
Redness and/or overheating of joints	4 / 109 (4%)	0 / 16 (0%)	3 / 65 (5%)	1 / 28 (4%)	0.679
Gastro-intestinal	NA	NA	NA	NA	NA
Stomach pain	65 / 109 (60%)	10 / 16 (62%)	37 / 65 (57%)	18 / 28 (64%)	0.777

Nausea	55 / 109 (50%)	10 / 16 (62%)	30 / 65 (46%)	15 / 28 (54%)	0.468
Loss of appetite	48 / 109 (44%)	11 / 16 (69%)	25 / 65 (38%)	12 / 28 (43%)	0.091
Diarrhoea	39 / 109 (36%)	7 / 16 (44%)	20 / 65 (31%)	12 / 28 (43%)	0.414
Flatulence	38 / 109 (35%)	8 / 16 (50%)	19 / 65 (29%)	11 / 28 (39%)	0.251
Bloatedness	29 / 109 (27%)	4 / 16 (25%)	13 / 65 (20%)	12 / 28 (43%)	0.072
Constipation	25 / 109 (23%)	5 / 16 (31%)	10 / 65 (15%)	10 / 28 (36%)	0.070
Weight loss	16 / 109 (15%)	2 / 16 (12%)	9 / 65 (14%)	5 / 28 (18%)	0.851
Vomiting	6 / 109 (6%)	1 / 16 (6%)	4 / 65 (6%)	1 / 28 (4%)	0.873
Body Temperature	NA	NA	NA	NA	NA
Increased sweating	63 / 109 (58%)	11 / 16 (69%)	39 / 65 (60%)	13 / 28 (46%)	0.301
Chills	34 / 109 (31%)	6 / 16 (38%)	18 / 65 (28%)	10 / 28 (36%)	0.627
Fever > 38.0 °C	18 / 109 (16%)	5 / 16 (31%)	8 / 65 (12%)	5 / 28 (18%)	0.184
Ears-Nose-Throat (ENT)	NA	NA	NA	NA	NA
Cold	46 / 109 (42%)	6 / 16 (38%)	29 / 65 (45%)	11 / 28 (39%)	0.820
Sore throat	40 / 109 (37%)	7 / 16 (44%)	24 / 65 (37%)	9 / 28 (32%)	0.743
Swollen lymph nodes	27 / 109 (25%)	6 / 16 (38%)	14 / 65 (22%)	7 / 28 (25%)	0.416
Dryness of the mouth	27 / 109 (25%)	2 / 16 (12%)	17 / 65 (26%)	8 / 28 (29%)	0.455
Altered taste	24 / 109 (22%)	5 / 16 (31%)	10 / 65 (15%)	9 / 28 (32%)	0.127
Problem swallowing	24 / 109 (22%)	5 / 16 (31%)	14 / 65 (22%)	5 / 28 (18%)	0.581
Ringing in ears	24 / 109 (22%)	4 / 16 (25%)	15 / 65 (23%)	5 / 28 (18%)	0.816
Altered smell	22 / 109 (20%)	3 / 16 (19%)	10 / 65 (15%)	9 / 28 (32%)	0.179
Ear pain	22 / 109 (20%)	3 / 16 (19%)	15 / 65 (23%)	4 / 28 (14%)	0.618
Sinus disorder	20 / 109 (18%)	1 / 16 (6%)	13 / 65 (20%)	6 / 28 (21%)	0.395
Voice disorder	18 / 109 (16%)	2 / 16 (12%)	12 / 65 (18%)	4 / 28 (14%)	0.792
Loss of smell	14 / 109 (13%)	0 / 16 (0%)	8 / 65 (12%)	6 / 28 (21%)	0.121
Hearing impairment	12 / 109 (11%)	3 / 16 (19%)	7 / 65 (11%)	2 / 28 (7%)	0.494
Loss of taste	9 / 109 (8%)	1 / 16 (6%)	4 / 65 (6%)	4 / 28 (14%)	0.405
Ophthalmic	NA	NA	NA	NA	NA
Light sensitivity	44 / 109 (40%)	7 / 16 (44%)	28 / 65 (43%)	9 / 28 (32%)	0.588
Dry eyes	29 / 109 (27%)	2 / 16 (12%)	17 / 65 (26%)	10 / 28 (36%)	0.243
Vision impairment	24 / 109 (22%)	5 / 16 (31%)	14 / 65 (22%)	5 / 28 (18%)	0.581

Conjunctivitis	24 / 109 (22%)	4 / 16 (25%)	13 / 65 (20%)	7 / 28 (25%)	0·826
Urogenital	NA	NA	NA	NA	NA
Painful menstruation	29 / 109 (27%)	0 / 16 (0%)	19 / 65 (29%)	10 / 28 (36%)	0·027
Problems passing urine	7 / 109 (6%)	1 / 16 (6%)	2 / 65 (3%)	4 / 28 (14%)	0·129
Changes in sexual function	1 / 109 (1%)	0 / 16 (0%)	0 / 65 (0%)	1 / 28 (4%)	0·232
Dermatological	NA	NA	NA	NA	NA
Hair loss	26 / 109 (24%)	0 / 16 (0%)	14 / 65 (22%)	12 / 28 (43%)	0·005
Skin rash	19 / 109 (17%)	3 / 16 (19%)	8 / 65 (12%)	8 / 28 (29%)	0·164
COVID-19 toes (purple, pink, bluish)	14 / 109 (13%)	0 / 16 (0%)	10 / 65 (15%)	4 / 28 (14%)	0·248
Other changes in skin color on hands and/or feet (red, white)	14 / 109 (13%)	2 / 16 (12%)	6 / 65 (9%)	6 / 28 (21%)	0·272
Swollen ankles / oedema	6 / 109 (6%)	0 / 16 (0%)	3 / 65 (5%)	3 / 28 (11%)	0·288

a Data are expressed as n/N (%) or median (IQR), as appropriate. Percentages may not total 100% because of rounding.

b P values were calculated using the χ^2 test or Kruskal-Wallis test, as appropriate.

c Abbreviations: MLCSQ, Munich Long COVID Symptom Questionnaire, PCC, Post-COVID Condition

NA = not available.

Table 3. Characteristics of Post-COVID Condition Assessed by Questionnaires and Defined by Specific ICD-10 Diagnoses.

	Overall^a	Children^a	Adolescents^a	Young Adults^a	P-Value^b
	N = 120	Age 7 to 11 years N = 16	Age 12 to 17 years N = 71	Age 18 to 25 years N = 33	
Participation	NA	NA	NA	NA	0.869
full participation	17 / 119 (14%)	1 / 16 (6%)	11 / 70 (16%)	5 / 33 (15%)	
> 50% participation	33 / 119 (28%)	7 / 16 (44%)	18 / 70 (26%)	8 / 33 (24%)	
< 50% participation	40 / 119 (34%)	5 / 16 (31%)	24 / 70 (34%)	11 / 33 (33%)	
no participation	29 / 119 (24%)	3 / 16 (19%)	17 / 70 (24%)	9 / 33 (27%)	
Bell CFIDS disability scale (Bell Score)	40.0 (30.0 - 60.0)	50.0 (40.0 - 60.0)	40.0 (30.0 - 50.0)	50.0 (30.0 - 60.0)	0.111
Fatigue Severity Scale (FSS)	6.4 (5.8 - 6.6)	6.4 (5.5 - 6.6)	6.4 (6.0 - 6.6)	6.3 (5.5 - 6.6)	0.656
DePaul University Symptom Questionnaire for Post-exertional Malaise (DSQ-PEM)	NA	NA	NA	NA	NA
PEM screening positive (DSQ-PEM)	112 / 119 (94%)	13 / 15 (87%)	66 / 71 (93%)	33 / 33 (100%)	0.113
PEM duration (hours)	NA	NA	NA	NA	0.476
< 1	5 / 114 (4%)	0 / 14 (0%)	4 / 70 (6%)	1 / 30 (3%)	
2 - 3	31 / 114 (27%)	3 / 14 (21%)	23 / 70 (33%)	5 / 30 (17%)	
4 - 10	28 / 114 (25%)	2 / 14 (14%)	16 / 70 (23%)	10 / 30 (33%)	
11 - 13	8 / 114 (7%)	1 / 14 (7%)	4 / 70 (6%)	3 / 30 (10%)	
14 - 23	19 / 114 (17%)	5 / 14 (36%)	8 / 70 (11%)	6 / 30 (20%)	
≥ 24	23 / 114 (20%)	3 / 14 (21%)	15 / 70 (21%)	5 / 30 (17%)	
Short Form-36 Health Survey (SF-36)	NA	NA	NA	NA	NA
Physical functioning	60.0 (45.0 - 75.0)	55.0 (50.0 - 87.5)	60.0 (40.0 - 75.0)	60.0 (50.0 - 75.0)	0.499
Role physical	0.0 (0.0 - 25.0)	25.0 (0.0 - 62.5)	0.0 (0.0 - 25.0)	0.0 (0.0 - 25.0)	0.067
Role emotional	100.0 (33.3 - 100.0)	66.7 (50.0 - 100.0)	100.0 (0.0 - 100.0)	66.7 (33.3 - 100.0)	0.597
Vitality	25.0 (10.0 - 35.0)	25.0 (10.0 - 50.0)	25.0 (15.0 - 33.8)	20.0 (10.0 - 30.0)	0.734
Mental health	64.0 (49.0 - 80.0)	80.0 (62.0 - 80.0)	66.5 (48.0 - 79.0)	56.0 (48.0 - 72.0)	0.151
Social functioning	50.0 (25.0 - 75.0)	62.5 (37.5 - 75.0)	50.0 (25.0 - 75.0)	50.0 (25.0 - 62.5)	0.656
Bodily pain	41.0 (22.0 - 70.0)	41.0 (27.0 - 81.0)	41.0 (22.0 - 71.5)	32.0 (31.0 - 62.0)	0.882

General health	32.7 (25.0 - 48.8)	35.0 (25.0 - 53.5)	32.5 (25.5 - 48.8)	31.2 (30.0 - 45.0)	0.939
Physical Component Summary (SF36-PCS)	31.0 (27.2 - 36.1)	32.1 (27.8 - 44.9)	30.0 (28.0 - 33.8)	32.2 (27.1 - 38.8)	0.655
Mental Component Summary (SF36-MCS)	44.2 (35.1 - 51.0)	41.9 (39.5 - 47.9)	47.2 (35.1 - 53.1)	40.1 (34.1 - 51.0)	0.646
Composite Autonomic Symptom Score (COMPASS 31)	NA	NA	NA	NA	NA
Orthostatic intolerance	20.0 (12.0 - 28.0)	18.0 (9.0 - 24.0)	20.0 (12.0 - 32.0)	16.0 (12.0 - 20.0)	0.178
Vasomotor	0.0 (0.0 - 0.0)	0.0 (0.0 - 1.9)	0.0 (0.0 - 0.0)	0.0 (0.0 - 0.0)	0.430
Secretomotor	2.1 (0.0 - 4.3)	2.1 (0.0 - 6.4)	0.0 (0.0 - 4.3)	4.3 (0.0 - 4.3)	0.186
Gastrointestinal	5.4 (1.8 - 7.1)	5.4 (1.8 - 5.4)	4.9 (1.8 - 7.1)	5.4 (2.7 - 9.8)	0.292
Bladder	0.0 (0.0 - 0.0)	0.0 (0.0 - 0.0)	0.0 (0.0 - 0.0)	0.0 (0.0 - 1.1)	0.019
Pupillomotor	1.3 (0.3 - 2.3)	0.7 (0.0 - 1.0)	1.7 (0.5 - 2.3)	1.6 (1.0 - 2.3)	0.016
Total Score (C31 Total Score)	27.8 (17.8 - 38.8)	20.0 (11.0 - 35.9)	28.7 (20.2 - 40.9)	27.5 (19.2 - 37.9)	0.382
Munich Berlin Symptom Questionnaire (MBSQ)	NA	NA	NA	NA	NA
Canadian Consensus Criteria (CCC)	22 / 116 (19%)	3 / 15 (20%)	10 / 69 (14%)	9 / 32 (28%)	0.694
Institute of Medicine (IOM) criteria	29 / 116 (25%)	3 / 15 (20%)	15 / 69 (22%)	11 / 32 (34%)	>0.999
Pediatric Case Definition (PCD-J)	10 / 84 (12%)	3 / 15 (20%)	7 / 69 (10%)	NA	0.374
Clinical Diagnostic Worksheet (CDW-R)	10 / 84 (12%)	3 / 15 (20%)	7 / 69 (10%)	NA	0.374
Diagnoses	NA	NA	NA	NA	NA
ME/CFS (ICD10-GM G93.3)	29 / 120 (24%)	3 / 16 (19%)	14 / 71 (20%)	12 / 33 (36%)	0.157
PoTS (ICD10-GM G90.8)	25 / 120 (21%)	3 / 16 (19%)	18 / 71 (25%)	4 / 33 (12%)	0.378

a Data are expressed as No. (%), median (IQR), or mean (SD), as indicated. Percentages may not total 100% because of rounding.

b P values were calculated using the χ^2 test, Fisher exact test, or Kruskal-Wallis test, depending on the distribution and group size.

c Abbreviations: ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome; PEM, post-exertional malaise; PoTS, postural orthostatic tachycardia syndrome

NA = not available.

Table 4. Risk Factor Analyses

	Unadj. Coefficient / Unadj. Log Odds Ratio^b	P-Value^a	Adj. Coefficient / Adj. Log Odds Ratio^b	P-Value^a
Bell Score^c	NA	NA	NA	NA
Sex, female	4.238	0.202	NA	NA
Age, years	0.431	0.292	NA	NA
Number of acute symptoms	-0.428	0.090	-0.439	0.081
COVID-19 vaccination prior infection	-0.537	0.882	0.266	0.942
Hospitalization due to COVID-19	-7.512	0.252	-8.991	0.176
Acute gastrointestinal symptoms ^d	-1.426	0.700	-0.201	0.958
Acute orthostatic symptoms ^e	-3.391	0.323	-3.524	0.304
Acute fatigue	7.511	0.177	7.446	0.180
Acute troubles concentrating	-5.595	0.133	-5.067	0.176
Infection susceptibility	-3.406	0.398	-3.549	0.378
Fatigue Severity Scale^c	NA	NA	NA	NA
Sex, female	-0.141	0.416	NA	NA
Age, years	0.004	0.862	NA	NA
Number of acute symptoms	0.045	<0.001	0.045	<0.001
COVID-19 vaccination prior infection	0.038	0.840	0.044	0.818
Hospitalization due to COVID-19	0.461	0.231	0.461	0.166
Acute gastrointestinal symptoms ^d	0.232	0.232	0.196	0.331
Acute orthostatic symptoms ^e	0.383	0.037	0.377	0.042
Acute fatigue	-0.040	0.895	-0.028	0.929
Acute troubles concentrating	0.522	0.008	0.502	0.013
Infection susceptibility	-0.214	0.314	-0.207	0.332
COMPASS 31 Total Score^c	NA	NA	NA	NA
Sex, female	8.626	0.007	NA	NA
Age, years	0.016	0.969	NA	NA
Number of acute symptoms	0.620	0.014	0.633	0.010
COVID-19 vaccination prior infection	1.792	0.619	1.334	0.706
Hospitalization due to COVID-19	8.373	0.171	4.249	0.492

Acute gastrointestinal symptoms ^d	-3.279	0.375	-0.773	0.838
Acute orthostatic symptoms ^e	12.346	<0.001	12.570	<0.001
Acute fatigue	-13.845	0.015	-14.791	0.007
Acute troubles concentrating	4.447	0.235	5.597	0.127
Infection susceptibility	-1.651	0.690	-1.897	0.635
SF36 – PCS^c	NA	NA	NA	NA
Sex, female	1.232	0.562	NA	NA
Age, years	-0.005	0.984	NA	NA
Number of acute symptoms	-0.514	0.001	-0.522	0.001
COVID-19 vaccination prior infection	-1.260	0.557	-1.329	0.557
Hospitalization due to COVID-19	-0.669	0.853	-1.111	0.764
Acute gastrointestinal symptoms ^d	-3.041	0.201	-2.818	0.259
Acute orthostatic symptoms ^e	-3.417	0.116	-3.492	0.112
Acute fatigue	-2.021	0.595	-2.115	0.581
Acute troubles concentrating	-6.317	0.007	-6.323	0.007
Infection susceptibility	-5.304	0.028	-5.335	0.029
SF36 – MCS^c	NA	NA	NA	NA
Sex, female	-4.469	0.054	NA	NA
Age, years	-0.978	0.730	NA	NA
Number of acute symptoms	-0.337	0.059	-0.335	0.060
COVID-19 vaccination prior infection	3.234	0.185	3.446	0.170
Hospitalization due to COVID-19	-8.990	0.020	-7.685	0.048
Acute gastrointestinal symptoms ^d	-1.435	0.583	-2.875	0.286
Acute orthostatic symptoms ^e	-6.344	0.007	-6.185	0.008
Acute fatigue	-4.152	0.317	-3.905	0.345
Acute troubles concentrating	-4.699	0.070	-4.788	0.063
Infection susceptibility	-1.417	0.599	-1.233	0.643
Number of Chronic Symptoms^c	NA	NA	NA	NA
Sex, female	4.074	0.089	NA	NA
Age, years	0.421	0.151	NA	NA
Number of acute symptoms	1.145	<0.001	1.142	<0.001

COVID-19 vaccination prior infection	-1.963	0.449	-1.303	0.618
Hospitalization due to COVID-19	6.107	0.208	4.212	0.394
Acute gastrointestinal symptoms ^d	8.422	<0.001	9.668	<0.001
Acute orthostatic symptoms ^e	9.314	<0.001	9.276	<0.001
Acute fatigue	4.536	0.341	3.802	0.425
Acute troubles concentrating	10.151	<0.001	10.900	<0.001
Infection susceptibility	4.284	0.144	4.121	0.158
ME/CFS Diagnosis (ICD-10 GM G93.3)^c	NA	NA	NA	NA
Sex, female	1.234	0.014	NA	NA
Age, years	0.068	0.206	NA	NA
Number of acute symptoms	0.010	0.765	0.010	0.762
COVID-19 vaccination prior infection	-0.692	0.206	-0.668	0.241
Hospitalization due to COVID-19	-1.030	0.342	-1.433	0.191
Acute gastrointestinal symptoms ^d	0.065	0.893	0.371	0.476
Acute orthostatic symptoms ^e	-0.423	0.341	-0.444	0.334
Acute fatigue	-0.923	0.144	-0.993	0.131
Acute troubles concentrating	0.375	0.469	0.553	0.302
Infection susceptibility	-0.256	0.647	-0.301	0.597
PoTS Diagnosis (ICD-10 GM G90.80)^c	NA	NA	NA	NA
Sex, female	0.372	0.426	NA	NA
Age, years	-0.055	0.355	NA	NA
Number of acute symptoms	0.068	0.071	0.074	0.051
COVID-19 vaccination prior infection	0.231	0.639	0.140	0.782
Hospitalization due to COVID-19	1.447	0.090	1.400	0.106
Acute gastrointestinal symptoms ^d	0.047	0.926	0.195	0.718
Acute orthostatic symptoms ^e	0.249	0.606	0.353	0.476
Acute fatigue	0.178	0.829	0.111	0.894
Acute troubles concentrating	0.454	0.414	0.607	0.290
Infection susceptibility	0.755	0.144	0.748	0.152

Brain Fog Diagnosis (ICD-10 GM F06.7)^c	NA	NA	NA	NA
Sex, female	-0.260	0.487	NA	NA
Age, years	0.075	0.118	NA	NA
Number of acute symptoms	0.005	0.850	0.002	0.942
COVID-19 vaccination prior infection	-0.245	0.549	-0.146	0.729
Hospitalization due to COVID-19	-0.954	0.193	-0.893	0.233
Acute gastrointestinal symptoms ^d	-0.066	0.874	-0.165	0.705
Acute orthostatic symptoms ^e	-0.420	0.288	-0.501	0.214
Acute fatigue	0.568	0.358	0.624	0.319
Acute troubles concentrating	0.503	0.234	0.441	0.308
Infection susceptibility	-0.522	0.248	-0.522	0.257
PEM^c	NA	NA	NA	NA
Sex, female	0.110	0.889	NA	NA
Age, years	0.339	0.013	NA	NA
Number of acute symptoms	0.160	0.021	0.152	0.033
COVID-19 vaccination prior infection	-1.661	0.063	-1.591	0.100
Hospitalization due to COVID-19	-0.744	0.514	-1.263	0.324
Acute gastrointestinal symptoms ^d	0.836	0.448	1.473	0.266
Acute OI symptoms ^e	0.939	0.235	0.819	0.330
Acute fatigue	1.366	0.129	2.125	0.047
Acute troubles concentrating	2.065	0.017	2.360	0.019
Infection susceptibility	0.654	0.554	0.593	0.605
PEM Duration^c	NA	NA	NA	NA
Sex, female	0.005	0.988	NA	NA
Age, years	-0.012	0.762	NA	NA
Number of acute symptoms	0.041	0.133	0.041	0.127
COVID-19 vaccination prior infection	-0.568	0.133	-0.623	0.104
Hospitalization due to COVID-19	-0.106	0.872	-0.136	0.839
Acute gastrointestinal symptoms ^d	0.270	0.467	0.317	0.406
Acute orthostatic symptoms ^e	-0.038	0.913	-0.033	0.927

Acute fatigue	-0.495	0.384	-0.506	0.376
Acute troubles concentrating	0.070	0.857	-0.069	0.870
Infection susceptibility	-0.069	0.870	-0.089	0.833

a P values calculated using Wald tests.

b Adjusted models include sex and age as covariates.

c Bell Score, FSS, COMPASS 31 Total Score, SF36-PCS, SF36-MCS, and number of chronic symptoms were modeled using multivariable linear regression. ME/CFS, PoTS, and PEM were modeled using multivariable logistic regression; PEM duration was modeled using proportional odds models.

d Gastrointestinal symptoms included vomiting, diarrhea, or gastrointestinal complaints.

e Orthostatic symptoms included dizziness, syncope, blackouts, or circulation dysregulation.

f Abbreviations: COVID-19, coronavirus disease 2019; FSS, Fatigue Severity Scale; GI, gastrointestinal; ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome; OI, orthostatic intolerance; PCS, Physical Component Summary; PEM, post-exertional malaise; PoTS, postural orthostatic tachycardia syndrome; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2; SF36-MCS, Short Form-36 Mental Component Summary; SF36-PCS, Short Form-36 Physical Component Summary.

NA = not available.